

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 4

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Recap of Previous Lecture

MLE for LVM

$$\sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i) = \sum_{i=1}^n \log \int p_{\theta}(\mathbf{x}_i|\mathbf{z}_i)p(\mathbf{z}_i)d\mathbf{z}_i \rightarrow \max_{\theta}.$$

Naive Monte Carlo Estimation

$$\begin{aligned} \log p_{\theta}(\mathbf{x}) &= \log \mathbb{E}_{p(\mathbf{z})} p_{\theta}(\mathbf{x}|\mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) \\ &\approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x}|\mathbf{z}_k), \quad \text{where } \mathbf{z}_k \sim p(\mathbf{z}). \end{aligned}$$

Variational Evidence Lower Bound (ELBO)

$$\begin{aligned} \mathcal{L}_{q,\theta}(\mathbf{x}) &= \mathbb{E}_q \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \\ \log p_{\theta}(\mathbf{x}) &= \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{q,\theta}(\mathbf{x}). \end{aligned}$$

Recap of Previous Lecture

Variational Evidence Lower Bound (ELBO)

$$\mathcal{L}_{q,\theta}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \leq \log p_{\theta}(\mathbf{x})$$

$$\mathcal{L}_{q,\theta}(\mathbf{x}) = \int q(\mathbf{z}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} d\mathbf{z} = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$$

Log-Likelihood Decomposition

$$\log p_{\theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z})) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})).$$

- Rather than maximizing likelihood, maximize the ELBO:

$$\max_{\theta} p_{\theta}(\mathbf{x}) \rightarrow \max_{q,\theta} \mathcal{L}_{q,\theta}(\mathbf{x})$$

- Maximizing the ELBO with respect to the variational distribution q is equivalent to minimizing the KL divergence:

$$\arg \max_q \mathcal{L}_{q,\theta}(\mathbf{x}) \equiv \arg \min_q D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})).$$

Recap of Previous Lecture

$$\begin{aligned}\mathcal{L}_{q,\theta}(\mathbf{x}) &= \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z})) \\ &= \mathbb{E}_q \left[\log p_{\theta}(\mathbf{x}|\mathbf{z}) - \log \frac{q(\mathbf{z})}{p(\mathbf{z})} \right] \rightarrow \max_{q,\theta}.\end{aligned}$$

Variational Posterior

$$q^*(\mathbf{z}) = \arg \max_q \mathcal{L}_{q,\theta^*}(\mathbf{x}) = \arg \min_q D_{\text{KL}}(q\|p) = p_{\theta^*}(\mathbf{z}|\mathbf{x}).$$

Amortized Variational Inference

We restrict the family of possible distributions $q(\mathbf{z})$ to a parametric class $q_{\phi}(\mathbf{z}|\mathbf{x})$, conditioned on data $\mathbf{x}$ and parameterized by ϕ .

Gradient Update

$$\begin{bmatrix} \phi_k \\ \theta_k \end{bmatrix} = \begin{bmatrix} \phi_{k-1} + \eta \cdot \nabla_{\phi} \mathcal{L}_{\phi,\theta}(\mathbf{x}) \\ \theta_{k-1} + \eta \cdot \nabla_{\theta} \mathcal{L}_{\phi,\theta}(\mathbf{x}) \end{bmatrix} \Big|_{(\phi_{k-1}, \theta_{k-1})}$$

Recap of Previous Lecture

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) \rightarrow \max_{\phi, \theta}.$$

Gradient $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \\ &\approx \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}^*), \quad \mathbf{z}^* \sim q_{\phi}(\mathbf{z}|\mathbf{x}). \end{aligned}$$

Gradient $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\begin{aligned} \nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \int p(\epsilon) \nabla_{\phi} \log p_{\theta}(\mathbf{x}|\mathbf{g}_{\phi}(\mathbf{x}, \epsilon)) d\epsilon - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) \\ &\approx \nabla_{\phi} \log p_{\theta}(\mathbf{x}|\mathbf{g}_{\phi}(\mathbf{x}, \epsilon^*)) - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) \end{aligned}$$

Variational Assumption

$$\begin{aligned} p(\epsilon) &= \mathcal{N}(0, \mathbf{I}); \quad q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\phi}(\mathbf{x}), \boldsymbol{\sigma}_{\phi}^2(\mathbf{x})). \\ \mathbf{z} &= \mathbf{g}_{\phi}(\mathbf{x}, \epsilon) = \boldsymbol{\sigma}_{\phi}(\mathbf{x}) \odot \epsilon + \boldsymbol{\mu}_{\phi}(\mathbf{x}). \end{aligned}$$

Recap of Previous Lecture

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\epsilon \sim p(\epsilon)$.
2. Reparametrize $\mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)$.
3. Compute the ELBO:

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) \approx \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})).$$

4. Update ϕ , θ via stochastic gradient ascent.

Sampling

1. Sample $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$.
2. Sample $\mathbf{x} \sim p_{\theta}(\mathbf{x}|\mathbf{z})$.

Note: The encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$ isn't needed during generation.

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- 01 ELBO Surgery and Optimal VAE Prior
- 02 Discrete VAE Latent Representations
- 03 Vector Quantized VAE (VQ-VAE)
- 04 Likelihood-Free Learning
- 05 Generative Adversarial Networks (GAN)

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ELBO Surgery

$$\frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi, \theta}(\mathbf{x}_i) = \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) \right].$$

Theorem

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg}, \phi}(\mathbf{z}) \| p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}];$$

- $q_{\text{agg}, \phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i)$ denotes the **aggregated** variational posterior.
- $\mathbb{I}_q[\mathbf{x}, \mathbf{z}]$ is the mutual information between $\mathbf{x}$ and $\mathbf{z}$ under the data distribution $p_{\text{data}}(\mathbf{x})$ and $q_{\phi}(\mathbf{z}|\mathbf{x})$.
- **The first term** encourages $q_{\text{agg}, \phi}(\mathbf{z})$ to match the prior $p(\mathbf{z})$.
- **The second term** reduces the information about $\mathbf{x}$ encoded in $\mathbf{z}$.

ELBO Surgery

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \| p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}].$$

Proof

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) &= \frac{1}{n} \sum_{i=1}^n \int q_{\phi}(\mathbf{z}|\mathbf{x}_i) \log \frac{q_{\phi}(\mathbf{z}|\mathbf{x}_i)}{p(\mathbf{z})} d\mathbf{z} \\ &= \frac{1}{n} \sum_{i=1}^n \int q_{\phi}(\mathbf{z}|\mathbf{x}_i) \log \frac{q_{\text{agg},\phi}(\mathbf{z}) q_{\phi}(\mathbf{z}|\mathbf{x}_i)}{p(\mathbf{z}) q_{\text{agg},\phi}(\mathbf{z})} d\mathbf{z} \\ &= \int \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i) \log \frac{q_{\text{agg},\phi}(\mathbf{z})}{p(\mathbf{z})} d\mathbf{z} + \frac{1}{n} \sum_{i=1}^n \int q_{\phi}(\mathbf{z}|\mathbf{x}_i) \log \frac{q_{\phi}(\mathbf{z}|\mathbf{x}_i)}{q_{\text{agg},\phi}(\mathbf{z})} d\mathbf{z} \\ &= D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \| p(\mathbf{z})) + \frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| q_{\text{agg},\phi}(\mathbf{z})) \\ \mathbb{I}_q[\mathbf{x}, \mathbf{z}] &= \frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| q_{\text{agg},\phi}(\mathbf{z})). \end{aligned}$$

ELBO Surgery

Revisiting the ELBO

$$\begin{aligned}
 \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi, \theta}(\mathbf{x}_i) &= \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) \right] \\
 &= \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} - \underbrace{\mathbb{I}_q[\mathbf{x}, \mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\text{KL}}(q_{\text{agg}, \phi}(\mathbf{z}) \| p(\mathbf{z}))}_{\text{Marginal KL}}
 \end{aligned}$$

The prior distribution $p(\mathbf{z})$ only appears in the last term.

Optimal VAE Prior

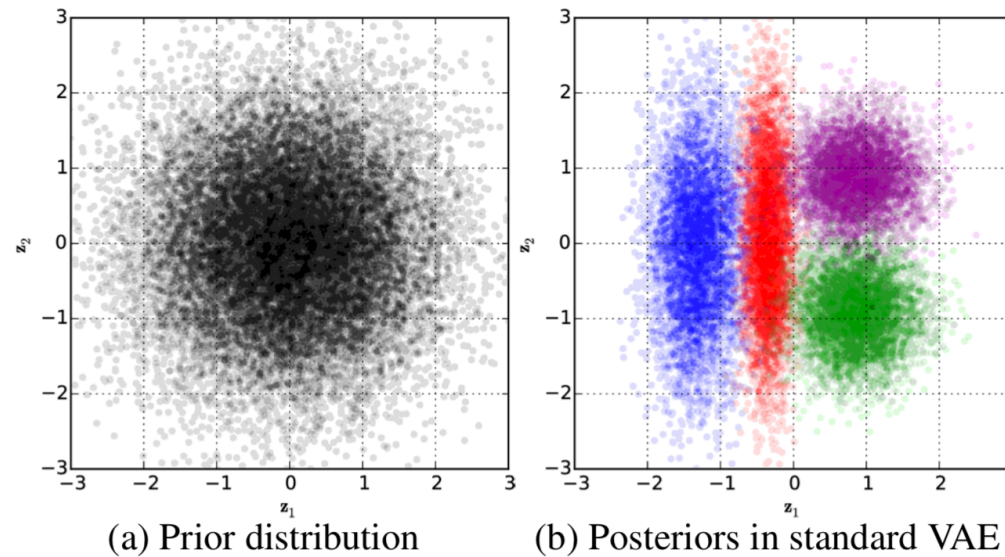
$$D_{\text{KL}}(q_{\text{agg}, \phi}(\mathbf{z}) \| p(\mathbf{z})) = 0 \quad \Leftrightarrow \quad p(\mathbf{z}) = q_{\text{agg}, \phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i).$$

Hence, the optimal prior $p(\mathbf{z})$ is the aggregated variational posterior $q_{\text{agg}, \phi}(\mathbf{z})$.

Marginal KL

$$D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z})\|p(\mathbf{z}))$$

- $q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\phi}(\mathbf{x}), \boldsymbol{\sigma}_{\phi}^2(\mathbf{x}))$ is unimodal.
- It is generally believed that the **mismatch between** $p(\mathbf{z})$ **and** $q_{\text{agg},\phi}(\mathbf{z})$ is the primary explanation for blurry VAE-generated images.



Why are VAE Generations Blurry?

Consider a Gaussian decoder $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_{\theta}(\mathbf{z}), \sigma^2 \mathbf{I})$.

ELBO Optimization

With **fixed** encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$, optimizing ELBO reduces to

$$\arg \min_{\theta} \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\phi}(\mathbf{z}|\mathbf{x})} \|\mathbf{x} - \boldsymbol{\mu}_{\theta}(\mathbf{z})\|^2$$

The optimal decoder mean is the **conditional expectation**:

$$\boldsymbol{\mu}^*(\mathbf{z}) = \mathbb{E}_{q_{\phi}(\mathbf{x}|\mathbf{z})}[\mathbf{x}] = \frac{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x})]}$$

Blurriness from Averaging

- $p_{\text{data}}(\mathbf{x})$ and $q_{\phi}(\mathbf{z}|\mathbf{x})$ give us the aggregated posterior $q_{\text{agg},\phi}(\mathbf{z})$.
- If $\mathbf{x} \neq \mathbf{x}'$ map to overlapping latent regions, $\boldsymbol{\mu}^*(\mathbf{z})$ averages over unrelated inputs.
- This leads to **blurry, non-distinct outputs**.

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Discrete VAE Latents

Motivation

- VAE has used **continuous** latent variables $\mathbf{z}$.
- For some modalities, **discrete** representations $\mathbf{z}$ may be a more natural choice.
- Advanced autoregressive models are highly effective for distributions over discrete variables (for example, transformers process discrete tokens).

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) \rightarrow \max_{\phi, \theta}.$$

- Apply the reparametrization trick to obtain unbiased gradients.
- Use Gaussian distributions for $q_{\phi}(\mathbf{z}|\mathbf{x})$ and $p(\mathbf{z})$ to compute the KL analytically.

Discrete VAE Latents

Assumptions

- Let $c \sim \text{Cat}(\boldsymbol{\pi})$, where

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^K \pi_k = 1.$$

- Suppose the VAE adopts a discrete latent variable c with prior $p(c) = \text{Uniform}\{1, \dots, K\}$.

ELBO

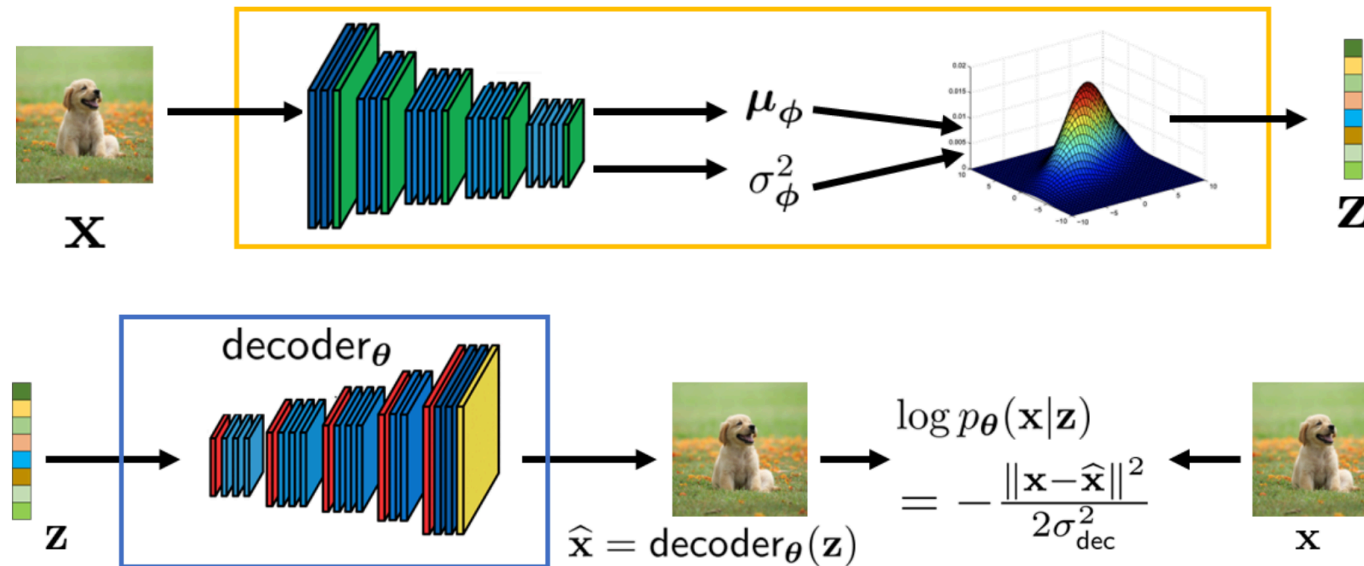
$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x}|c) - D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \| p(c)) \rightarrow \max_{\phi, \theta}.$$

$$\begin{aligned} D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \| p(c)) &= \sum_{k=1}^K q_{\phi}(k|\mathbf{x}) \log \frac{q_{\phi}(k|\mathbf{x})}{p(k)} \\ &= \sum_{k=1}^K q_{\phi}(k|\mathbf{x}) \log q_{\phi}(k|\mathbf{x}) - \sum_{k=1}^K q_{\phi}(k|\mathbf{x}) \log p(k) \\ &= -H(q_{\phi}(c|\mathbf{x})) + \log K. \end{aligned}$$

Discrete VAE Latents

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x}|c) + H(q_{\phi}(c|\mathbf{x})) - \log K \rightarrow \max_{\phi, \theta}.$$

- The encoder should output a discrete distribution $q_{\phi}(c|\mathbf{x})$.
- We need an analogue of the reparametrization trick for discrete $q_{\phi}(c|\mathbf{x})$.
- The decoder $p_{\theta}(\mathbf{x}|c)$ must take a discrete random variable c as input.



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Vector Quantization

Define the codebook (dictionary) space $\{\mathbf{e}_k\}_{k=1}^K$ with $\mathbf{e}_k \in \mathbb{R}^L$ and K the number of codebook entries.

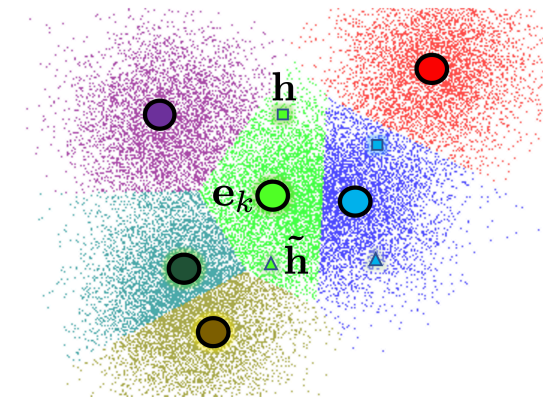
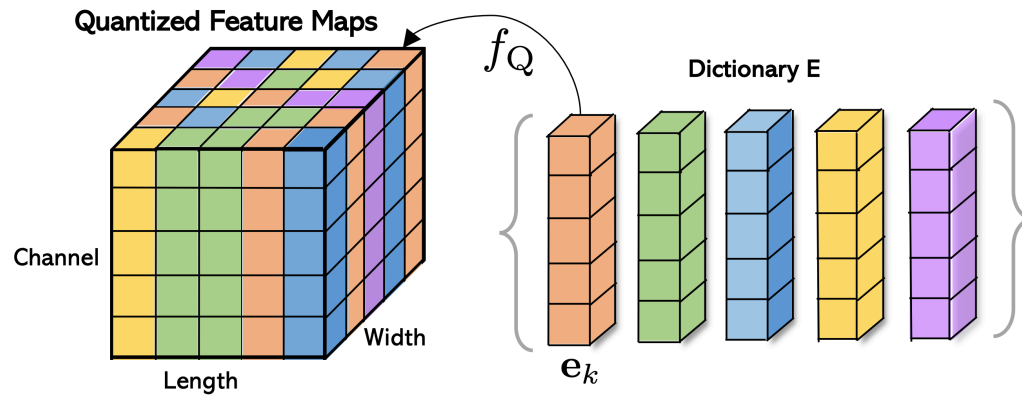
Quantized Representation

A quantized vector $\mathbf{z}_q \in \mathbb{R}^L$, for any $\mathbf{z} \in \mathbb{R}^L$, is defined via nearest-neighbor lookup in the codebook:

$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg \min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

Quantization Procedure

If the encoded tensor has spatial dimensions, quantization is independently applied to each of the $W \times H$ locations.



Vector Quantized VAE (VQ-VAE)

- The encoder outputs a continuous vector $\mathbf{z}_e = \text{NN}_{e,\phi}(\mathbf{x}) \in \mathbb{R}^L$.
- Quantization deterministically maps $\mathbf{z}_e$ to its quantized codebook vector $\mathbf{z}_q$.
- The decoder is conditioned on codebook entries $\mathbf{e}_c$, i.e., via $p_\theta(\mathbf{x}|\mathbf{e}_c)$ (instead of $p_\theta(\mathbf{x}|c)$).

Deterministic Variational Posterior

$$q_\phi(c = k^*|\mathbf{x}) = \begin{cases} 1, & \text{for } k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

$$D_{\text{KL}}(q_\phi(c|\mathbf{x})\|p(c)) = - \underbrace{\text{H}(q_\phi(c|\mathbf{x}))}_{=0} + \log K = \log K.$$

Note: The KL regularizer becomes constant and has no direct effect on the ELBO objective in this case.

Vector Quantized VAE (VQ-VAE): Forward

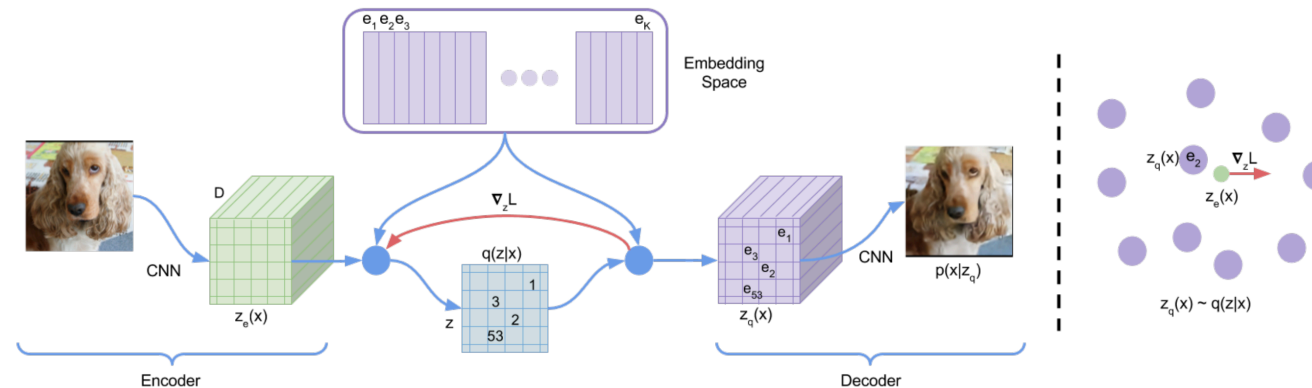
Deterministic Variational Posterior

$$q_{\phi}(c = k^* | \mathbf{x}) = \begin{cases} 1, & \text{for } k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x} | \mathbf{e}_c) - \log K = \log p_{\theta}(\mathbf{x} | \mathbf{z}_q) - \log K,$$

where $\mathbf{z}_q = \mathbf{e}_{k^*}$, $k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|$.

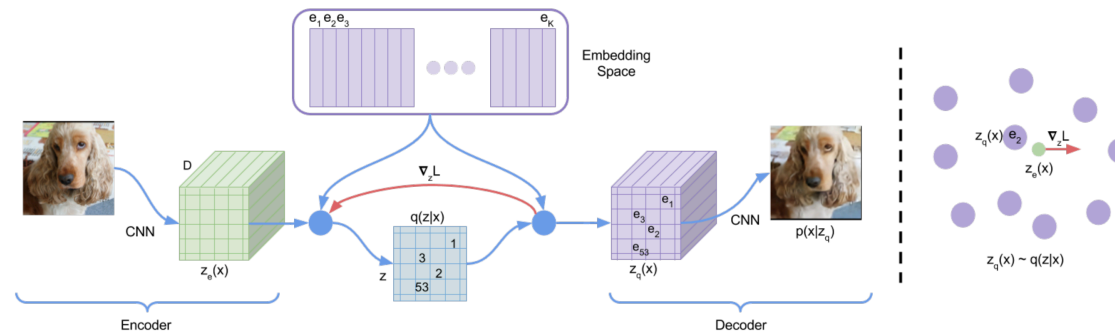


Challenge: The $\arg \min$ operation is non-differentiable.

Vector Quantized VAE (VQ-VAE): Backward

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \log p_{\theta}(\mathbf{x} | \mathbf{z}_q) - \log K, \quad \mathbf{z}_q = \mathbf{e}_{k^*}, \quad k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|.$$



Straight-Through Gradient Estimator

$$\begin{aligned} \frac{\partial \log p(\mathbf{x} | \mathbf{z}_q, \boldsymbol{\theta})}{\partial \boldsymbol{\phi}} &= \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \boldsymbol{\phi}} \\ &= \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \mathbf{z}_e} \cdot \frac{\partial \mathbf{z}_e}{\partial \boldsymbol{\phi}} \\ &\approx \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \boldsymbol{\phi}} \end{aligned}$$

Vector Quantized VAE-2 (VQ-VAE-2)

Extension to the spatial domain: $\mathbf{c} \in \{1, \dots, K\}^{W \times H}$

$$q_{\phi}(\mathbf{c}|\mathbf{x}) = \prod_{i=1}^W \prod_{j=1}^H q(c_{ij}|\mathbf{x}, \phi); \quad p(\mathbf{c}) = \prod_{i=1}^W \prod_{j=1}^H \text{Uniform}\{1, \dots, K\}.$$

Sample Diversity



VQ-VAE (Proposed)

BigGAN deep

Vector Quantized VAE (VQ-VAE): Final algorithm

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.
2. Compute the encoding $\mathbf{z}_e = \text{NN}_{e,\phi}(\mathbf{x})$.
3. Compute the quantized representation (per spatial location if applicable):

$$k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|_2, \quad \mathbf{z}_q = \mathbf{e}_{k^*}.$$

4. Compute the ELBO:

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \log p_{\theta}(\mathbf{x}|\mathbf{z}_q) - \log K.$$

5. Compute total loss with codebook and commitment terms:

$$\begin{aligned} \mathcal{L} = & -\mathcal{L}_{\phi,\theta}(\mathbf{x}) + \|\text{sg}[\mathbf{z}_e] - \mathbf{e}_{k^*}\|_2^2 \\ & + \beta \|\mathbf{z}_e - \text{sg}[\mathbf{e}_{k^*}]\|_2^2. \end{aligned}$$

6. Update ϕ, θ (straight-through estimator for the encoder).

Sampling

1. Sample $c \sim p(c)$
 $= \text{Uniform}\{1, \dots, K\}.$
2. Sample $\mathbf{x} \sim p_{\theta}(\mathbf{x}|\mathbf{e}_c).$

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Likelihood-Based Models

Poor Likelihood High-Quality Samples

$$p_1(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \mathcal{N}(\mathbf{x} | \mathbf{x}_i, \epsilon \mathbf{I})$$

If ϵ is very small, this model produces excellent, sharp samples but achieves poor likelihoods on test data.

High Likelihood Poor Samples

$$\begin{aligned} p_2(\mathbf{x}) &= 0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x}) \\ \log [0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x})] \\ &\geq \log[0.01p(\mathbf{x})] \\ &= \log p(\mathbf{x}) - \log 100 \end{aligned}$$

This model contains mostly noisy, irrelevant samples; for high dimensions, $\log p(\mathbf{x})$ scales linearly with m .

- Likelihood isn't always a suitable metric for evaluating generative models.
- Sometimes, the likelihood function can't even be computed exactly.

Likelihood-Free Learning

Motivation

We're interested in approximating the true data distribution $p_{\text{data}}(\mathbf{x})$. Instead of searching over all distributions, let's learn a model $p_{\theta}(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$.

Suppose we have two sets of samples:

- $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ — real data;
- $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\theta}(\mathbf{x})$ — generated (fake) data.

Define a discriminative model (classifier):

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\theta}(\mathbf{x}))$$

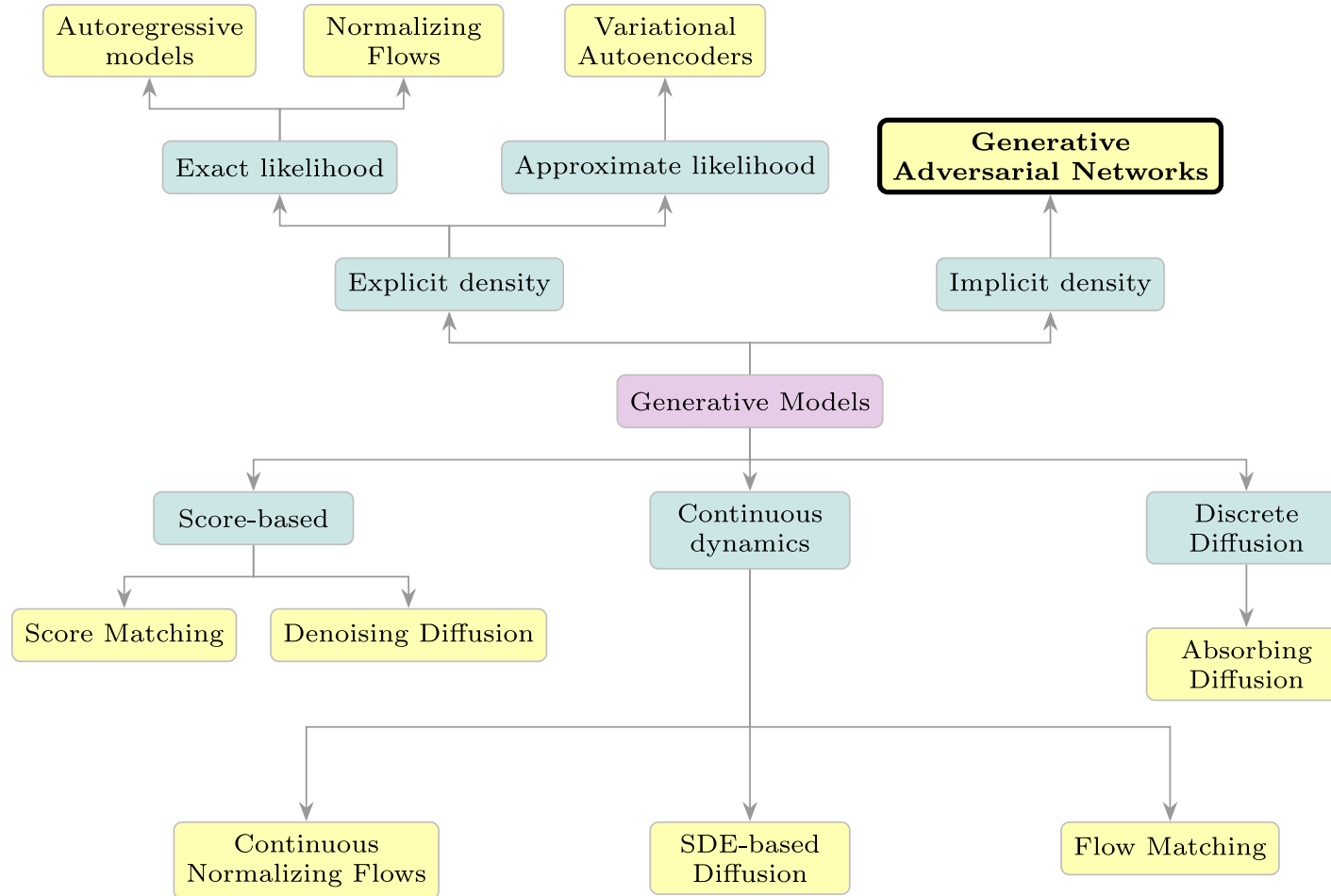
Assumption

The generative model $p_{\theta}(\mathbf{x})$ matches $p_{\text{data}}(\mathbf{x})$ if a discriminative model $p(y|\mathbf{x})$ can't distinguish between them — that is, if $p(y = 1|\mathbf{x}) = 0.5$ for every $\mathbf{x}$.

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Generative Models Taxonomy



Generative Adversarial Networks (GAN)

- The more expressive the discriminator, the closer we get to the optimal $p_\theta(\mathbf{x})$.
- Standard classifiers are trained by minimizing cross-entropy loss $-\mathbb{E}_{\hat{p}(\mathbf{x}, y)} \log p(y|\mathbf{x})$ with $\hat{p}(\mathbf{x}, y) = \frac{1}{2} \mathbb{I}_{y=1} p_{\text{data}}(\mathbf{x}) + \frac{1}{2} \mathbb{I}_{y=0} p_\theta(\mathbf{x})$.

Cross-Entropy for Discriminator

$$\min_{p(y|\mathbf{x})} \left[-\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) - \mathbb{E}_{p_\theta(\mathbf{x})} \log p(y=0|\mathbf{x}) \right]$$

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) + \mathbb{E}_{p_\theta(\mathbf{x})} \log p(y=0|\mathbf{x}) \right]$$

Generative Model

Suppose $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})$, where $p(\mathbf{z})$ is a base distribution, and $p_\theta(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_\theta(\mathbf{z}))$ is deterministic.

Generative Adversarial Networks (GAN)

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log p(y = 0|\mathbf{x}) \right]$$

- **Discriminator:** A classifier $p_{\phi}(y = 1|\mathbf{x}) = D_{\phi}(\mathbf{x}) \in [0, 1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \right]$$

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

GAN Optimality

Theorem

The minimax game

$$\min_G \max_D \underbrace{\left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]}_{V(G,D)}$$

achieves its global optimum when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Proof (Fixed G)

$$\begin{aligned} V(G, D) &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \\ &= \int \underbrace{[p_{\text{data}}(\mathbf{x}) \log D(\mathbf{x}) + p_{\theta}(\mathbf{x}) \log(1 - D(\mathbf{x}))]}_{y(D)} d\mathbf{x} \\ \frac{dy(D)}{dD} &= \frac{p_{\text{data}}(\mathbf{x})}{D(\mathbf{x})} - \frac{p_{\theta}(\mathbf{x})}{1 - D(\mathbf{x})} = 0 \quad \Rightarrow \quad D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \end{aligned}$$

GAN Optimality

Proof Continued (Fixed $D = D^*$)

$$\begin{aligned}
 V(G, D^*) &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log \left(\frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right) \\
 &\quad + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log \left(\frac{p_{\theta}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right) \\
 &= D_{\text{KL}} \left(p_{\text{data}}(\mathbf{x}) \parallel \frac{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})}{2} \right) + D_{\text{KL}} \left(p_{\theta}(\mathbf{x}) \parallel \frac{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})}{2} \right) - 2 \log 2 \\
 &= 2 D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) - 2 \log 2.
 \end{aligned}$$

Jensen-Shannon Divergence (Symmetric KL Divergence)

$$D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) = \frac{1}{2} [D_{\text{KL}}(p_{\text{data}}(\mathbf{x}) \parallel \star) + D_{\text{KL}}(p_{\theta}(\mathbf{x}) \parallel \star)]$$

This can be regarded as a proper distance metric!

$$V(G^*, D^*) = -2 \log 2, \quad p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x}), \quad D^*(\mathbf{x}) = 0.5.$$

GAN Optimality

Theorem

The following minimax game

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

Reality

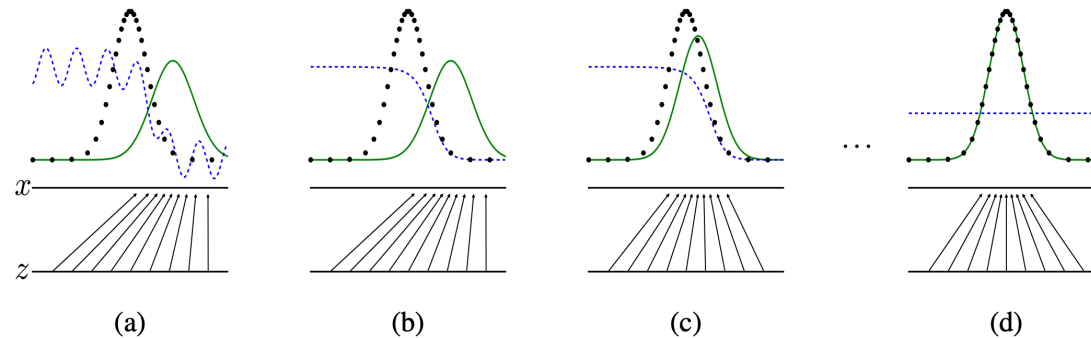
- Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- Generator and discriminator losses typically oscillate during GAN training.

GAN Training

Assume both generator and discriminator are parametric models: $D_\phi(\mathbf{x})$ and $\mathbf{G}_\theta(\mathbf{z})$.

Objective

$$\min_{\theta} \max_{\phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_\phi(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_\phi(\mathbf{G}_\theta(\mathbf{z}))) \right]$$



- $\mathbf{z} \sim p(\mathbf{z})$ is a latent variable.
- $p_\theta(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_\theta(\mathbf{z}))$ serves as a deterministic decoder (like normalizing flows).
- There is no encoder present.

Summary

- ELBO surgery gives insights into the prior's influence in VAEs; the optimal prior is the aggregated variational posterior.
- The mismatch between $p(\mathbf{z})$ and $q_{\text{agg},\phi}(\mathbf{z})$ is widely regarded as the principal reason for VAE-generated image blurriness.
- Vector quantization provides a way to construct VAEs with discrete latents and deterministic variational posteriors.
- The straight-through gradient estimator allows gradients to pass as if quantization were an identity operation during backpropagation.
- Likelihood is not always a suitable metric for evaluating generative models; likelihood-free learning motivates the use of discriminative models to match distributions.
- GANs optimize the Jensen-Shannon divergence in theory; training alternates generator and discriminator updates to match the data distribution.